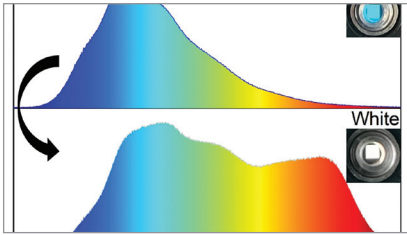


Simplified white OLED design enables flexible, low-cost panels

Researchers at the University of Turku (Turku, Finland) have developed a simplified white organic light-emitting



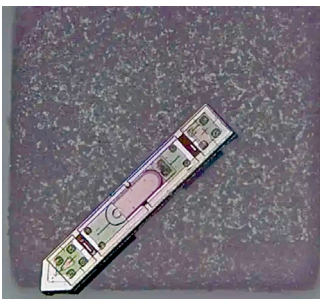
The image displays the electroluminescence spectra of the fabricated device, demonstrating the colour conversion from blue to white emission (Credit: www.utu.fi/en)

diode (OLED) design that removes costly RGB dopants and indium tin oxide electrodes. Using a single blue TADF emitter within an aluminium microcavity, the device produces tunable white

light between 3790K and 5050K. The approach cuts material complexity and rare-metal use while enabling flexible, low-cost OLED panels for sustainable lighting and smart-building applications.

Microscale implant enables long-term, minimally invasive brain monitoring

Researchers at Cornell University (Ithaca, New York) have developed a brain implant smaller than a grain of salt that can wirelessly record neural activity for over a year.



Cornell-developed neural implant small enough to sit atop a grain of salt (Credit: <https://interestingengineering.com>)

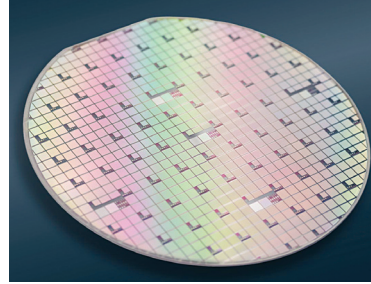
Known as the microscale optoelectronic tetherless electrode (MOTE), it uses infrared light to both power itself and transmit data, eliminating the need for wires or batteries. Built from aluminium gallium arsenide, the device enables long-term, minimally invasive brain monitoring for future neurotech and medical applications.

Electronic fibre keeps conducting even if stretched ten times its length

Researchers at École Polytechnique Fédérale de Lausanne, EPFL (Lausanne, Switzerland) have created an 'electronic fibre' that stays conductive even when stretched over ten times its length. Made by embedding indium-gallium liquid metal droplets in a soft elastomer and drawn using fibre-optic techniques, the fibre forms controllable conductive paths as it stretches. Tested in a knee brace, it accurately tracked motion in real time, pointing to future applications in smart clothing, soft robotics, and motion-sensing textiles.

Vertical GaN could reshape energy management in AI data centres

Researchers at Onsemi (Phoenix, Arizona, United States) have unveiled a vertical gallium nitride (vGaN) power



Onsemi's vertical GaN wafer manufactured in Syracuse, NY (Credit: Onsemi)

technology that could reshape how energy is managed in AI data centres, electric vehicles, and renewable systems. By allowing current to flow vertically through GaN-on-GaN substrates, the design cuts energy losses by up to 50% and triples

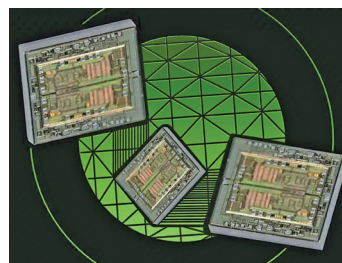
power density compared to lateral GaN devices. The result: smaller, cooler, and more efficient power electronics across high-demand industries.

3D porous electrode design for EV batteries

Researchers from POSTECH (Pohang, South Korea) and Chung-Ang University (Seoul, South Korea) have developed a 3D porous electrode that could make lithium-metal batteries safer and more powerful for electric vehicles. The design uses vertically aligned channels with a gradient of lithium affinity, guiding uniform lithium deposition and preventing dendrite growth, which is a key cause of battery fires. Delivering 398Wh/kg and 1516Wh/L, the batteries outperform today's lithium-ion cells, offering a scalable path toward safer, longer-range EVs.

Thin-film analogue IC for smaller and faster electronics

Ok Electric Industry (Minato City, Tokyo, Japan) and Nisshinbo Micro Devices (Tokyo, Japan) are co-developing ultra-thin



Thin-film 3D analog IC developed by Ok Electric Industry Co. and Nisshinbo Micro Devices could find applications in AI, IoT devices, and autonomous vehicles (Credit: <https://spectrum.ieee.org>)

3D analogue ICs using crystal film bonding (CFB) to shrink and simplify chip production. The technique stacks 5-10µm chip layers joined by molecular forces and separated by silicon oxide, enabling low-cost fabrication with standard tools. Aluminium shielding minimises cross-talk between layers, while modular chiplets enhance

flexibility. The partners aim to commercialise the technology by 2026 for AI, IoT, and automotive systems.